IGCSE 0607 Practice Questions — E7.5 — Combined Questions & Markscheme

Non-right-angled triangles

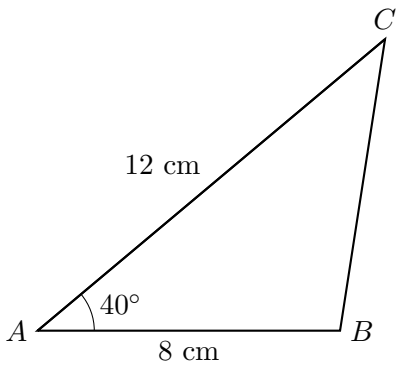
Total Questions: 6 — **Total Marks:** 40 Marks

Table of Contents & Quick Navigation

- **E7.5** Non-right-angled triangles
  → **Paper 4 (Calculator)** *(6 Qs)*

---

1 The diagram shows a triangle ABC.



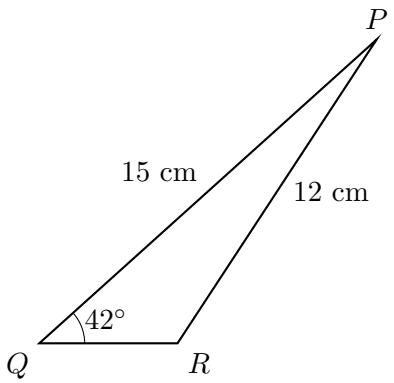
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

2 The diagram shows a triangle PQR.
The length of PQ is 15 cm and the length of PR is 12 cm.
Angle $PQR = 42°$.
Angle PRQ is obtuse.



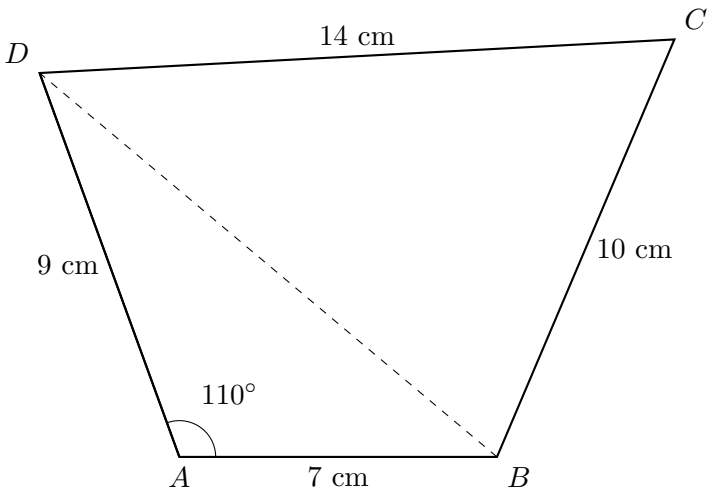
(a) Calculate angle PRQ.

..... ° **[3]**

(b) Calculate the length of QR.

..... cm **[3]**

3 The diagram shows a quadrilateral $ABCD$.
$AB = 7$ cm, $AD = 9$ cm, $BC = 10$ cm and $CD = 14$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of BD.

..... cm **[3]**

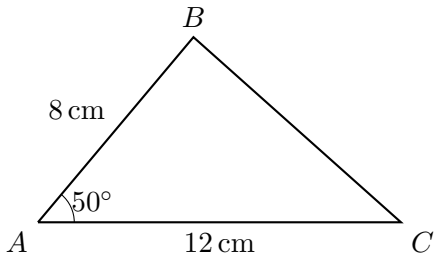
(b) Calculate angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

4 A triangle ABC is shown below.



NOT TO SCALE

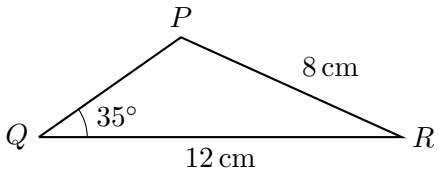
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

5 In triangle PQR, $PR = 8$ cm, $QR = 12$ cm and angle $PQR = 35°$.
Angle QPR is an obtuse angle.



NOT TO SCALE

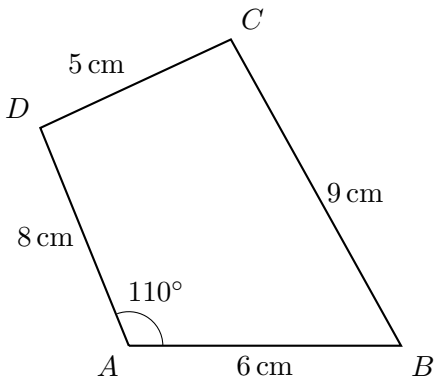
(a) Calculate the size of angle QPR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

6 The diagram shows a quadrilateral $ABCD$.
$AB = 6$ cm, $AD = 8$ cm, $DC = 5$ cm, and $BC = 9$ cm.
Angle $DAB = 110°$.



NOT TO SCALE

(a) Calculate the length of the diagonal BD.

..... *cm* **[3]**

(b) Calculate the size of angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

1 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin 40°$ **[M1]**

30.9 (accept 30.85 . . .) **[A1]**

(b) $8^2 + 12^2 - 2 \times 8 \times 12 \times \cos 40°$ **[M1]**

60.9 . . . **[A1]**

7.80 or 7.81 **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** $\frac{\sin R}{15} = \frac{\sin 42°}{12}$ **[M1]**

$\sin R = \frac{15 \sin 42°}{12}$ (evaluates to 0.836 . . .) **[M1]**

123.2 (accept 123.2 to 123.3) **[A1]**

(b) Angle $P = 180° - 42° - 123.24° = 14.76° \ldots$ **[M1]**

$\frac{QR}{\sin 14.76°} = \frac{12}{\sin 42°}$ (or equivalent application of sine/cosine rule) **[M1]**

4.57 (accept 4.56 to 4.58) **[A1]**

[Back to Question] *[Total Marks: 6]*

3 **(a)** $7^2 + 9^2 - 2 \times 7 \times 9 \times \cos 110°$ **[M1]**

173.09 . . . **[A1]**

13.2 (accept 13.15 to 13.2) **[A1]**

(b) $13.156^2 = 10^2 + 14^2 - 2 \times 10 \times 14 \times \cos(\angle BCD)$ **[M1]**

$\cos(\angle BCD) = \frac{10^2 + 14^2 - 173.09}{2 \times 10 \times 14}$ **[M1]**

64.0 (accept 63.9 . . . to 64.0) **[A1]**

(c) $\frac{1}{2} \times 7 \times 9 \times \sin 110°$ (evaluates to 29.6 . . .) **[M1]**

$\frac{1}{2} \times 10 \times 14 \times \sin 64.0°$ (evaluates to 62.9 . . .) **[M1]**

92.5 (accept 92.4 to 92.6) **[A1]**

[Back to Question] *[Total Marks: 9]*

4 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin(50°)$ **[M1]**

36.77... so 36.8 **[A1]**

(b) $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(50°)$ **[M1]**

$BC^2 = 208 - 123.415... = 84.58...$ **[M1]**

9.197... so 9.20 **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $\dfrac{\sin P}{12} = \dfrac{\sin 35^\circ}{8}$ **[M1]**

$\sin P = \dfrac{12 \sin 35^\circ}{8} = 0.8603...$ **[M1]**

Acute angle $= 59.35...^\circ$ **[A1]**

Obtuse angle $P = 180^\circ - 59.35...^\circ = 120.64...^\circ$ so 120.6° **[A1]**

(b) Angle $R = 180^\circ - 35^\circ - 120.64...^\circ = 24.36...^\circ$ **[M1]**

Area $= \dfrac{1}{2} \times 8 \times 12 \times \sin(24.36...^\circ) = 19.79...$ so 19.8 **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** $BD^2 = 6^2 + 8^2 - 2(6)(8)\cos(110^\circ)$ **[M1]**

$BD^2 = 100 - 96(-0.342...) = 132.83...$ **[M1]**

$BD = 11.52...$ so 11.5 **[A1]**

(b) $\cos C = \dfrac{9^2 + 5^2 - (11.525...)^2}{2(9)(5)}$ **[M1]**

$\cos C = \dfrac{106 - 132.83...}{90} = -0.2981...$ **[M1]**

$C = 107.34...^\circ$ so 107.3° **[A1]**

(c) Area of triangle $ABD = \dfrac{1}{2} \times 6 \times 8 \times \sin(110^\circ) = 22.55...$ **[M1]**

Area of triangle $BCD = \dfrac{1}{2} \times 9 \times 5 \times \sin(107.34...^\circ) = 21.47...$ **[M1]**

Total Area $= 22.55... + 21.47... = 44.02...$ so 44.0 **[A1]**

[Back to Question] *[Total Marks: 9]*